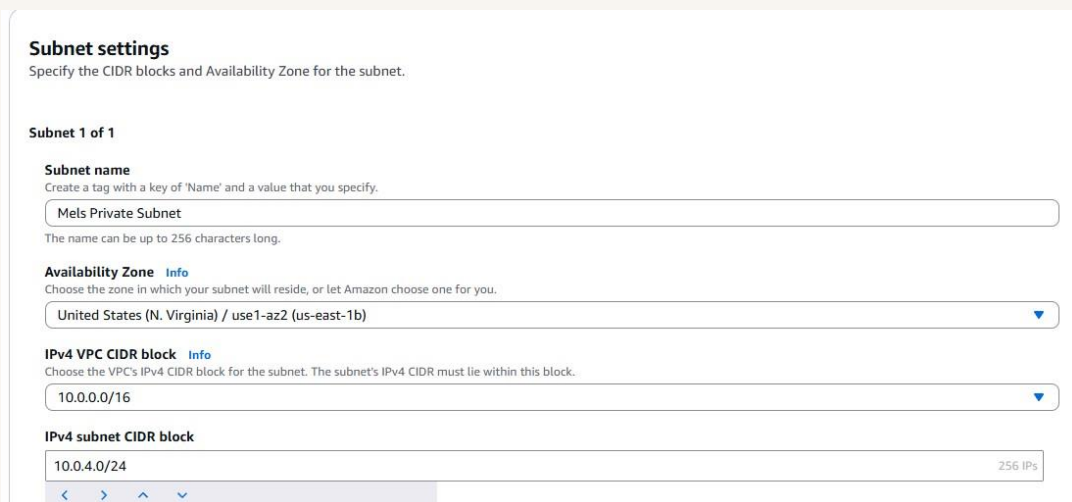


Creating a Private Subnet

Melvin green



Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a service that allows you to create a logically isolated network within AWS where you can deploy resources such as EC2 instances and databases. It

is useful because it gives you full control over your network configuration, including the ability to create public and private subnets, route tables, internet gateways, and Network ACLs. Allowing you to securely separate public-facing resources from confidential data and internal services.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a private subnet and private route table, separating them from the public subnet and route table, while ensuring that no access to the private subnet was possible through the internet gateway.

One thing I didn't expect in this project was...

One thing I did not expect in this project was that when creating a new Network ACL, it will have the default ACL rules until they are manually changed.

This project took me...

This project took me roughly 45 minutes. I am starting to understand the AWS dashboard better, and it is becoming easier to maneuver.

I am recreating my VPC, subnets, internet gateways, and ACLs repeatedly for more practice, while also learning new concepts such as creating a private subnet and route table, and ensuring that no access is permitted from any outside sources.

Private vs Public Subnets

The difference between public and private subnets is that public subnets are accessible by and can access the internet, while private subnets are completely isolated from the internet by default.

Having private subnets is useful because keeping resources away from the internet is extremely important for securing confidential resources and data.

Private and public subnets cannot have the same IPv4 CIDR block (i.e Private and public subnets cannot have the same IPv4 CIDR block (i.e., the same range of IP addresses). The CIDR block for every subnet must be unique and cannot overlap with another subnet.

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Mels Private Subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az2 (us-east-1b) ▼

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.4.0/24 256 IPs

< > ^ v

A dedicated route table

By default, my private subnet is associated with the default route table, i.e., a route table that has a route to an internet gateway

I had to set up a new route table because my subnet can't have a route to the internet gateway

My private subnet's dedicated route table only has one inbound and one outbound rule that allows internal communication, i.e., with a destination of another resource within my VPC

rtb-0f2b107a0fd359959 / Mel's Private Route Table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

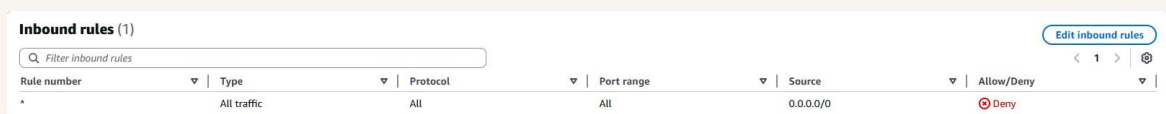
Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No

A new network ACL

By default, my private subnet is associated with the default network ACL that's set up for every VPC created in my AWS account.

I set up a dedicated network ACL for my private subnet because a network ACL becomes crucial in the event of security breaches, where traffic that has compromised my public subnet can get access to my private subnet if I have network ACL rules that allow all inbound/outbound traffic.

My new network ACL has two simple rules - deny all inbound and deny all outbound traffic



The screenshot shows the AWS Management Console interface for configuring Network ACL (NACL) inbound rules. The title is 'Inbound rules (1)'. There is a search bar labeled 'Filter inbound rules'. Below the search bar is a table with columns: Rule number, Type, Protocol, Port range, Source, and Allow/Deny. The table contains one rule with the following details: Rule number is '*', Type is 'All traffic', Protocol is 'All', Port range is 'All', Source is '0.0.0.0/0', and Allow/Deny is 'Deny' (indicated by a red circle with a slash). There is an 'Edit inbound rules' button in the top right corner.

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny